Year	Q1	Q2	Q3	Q4
2017	76	68	67	95
2016	75	66	64	89
2015	72	64	63	75

The seasonal index

A **season index** is defined by the formula:

$$\text{seasonal index} = \frac{\text{value for season}}{\text{seasonal average}}$$

In this formula, the season is a month, quarter or the like.

72/[(72+64+63+75)/4]=1.05109 (**the seasonal index** for 2015 Q1)

64/[(72+64+63+75)/4]=0.93430 (**the seasonal index** for 2015 Q2)

How to calculate the deseasonalized data based on three years?

Deseasonalized data = actual data / average seasonal index

Ave_SI_Q1 = (SI_2015_Q1+ SI_2016_Q1+ SI_2017_Q1)/3

= (1.051094891 + 1.020408163 + 0.993464052)/3 = 1.021655702

Deseasonalized (2015 Q1) = 72 / Ave_SI_Q1 = 70.5

Deseasonalized (2016 Q1) = 75 / Ave_SI_Q1 = 73.4

Deseasonalized (2017 Q1) = 76 / Ave_SI_Q1 = 74.4

Quarter	Actual	Seasonal Index	Deseasonalized Data
2015Q1	72	1.051094891	70.5
2015Q2	64	0.934306569	70.6
2015Q3	63	0.919708029	70.9
2015Q4	75	1.094890511	63.4
2016Q1	75	1.020408163	73.4
2016Q2	66	0.897959184	72.8
2016Q3	64	0.870748299	72.0
2016Q4	89	1.210884354	75.3
2017Q1	76	0.993464052	74.4
2017Q2	68	0.888888889	75.0
2017Q3	67	0.875816993	75.4
2017Q4	95	1.241830065	80.3